

0 6

Myelodysplastic syndromes (MDS) are a group of malignant cancers. In MDS, the bone marrow does not produce healthy blood cells.

Haematopoietic stem cell transplantation (HSCT) is one treatment for MDS. In HSCT, the patient receives stem cells from the bone marrow of a person who does not have MDS. Before the treatment starts, the patient's faulty bone marrow is destroyed.

0 6**1**

For some patients, HSCT is an effective treatment for MDS.

Explain how.

[3 marks]

0 6**2**

MDS can develop from epigenetic changes to tumour suppressor genes. In some patients, the drug AZA has reduced the effects of MDS. AZA is an inhibitor of DNA methyltransferases. These enzymes add methyl groups to cytosine bases.

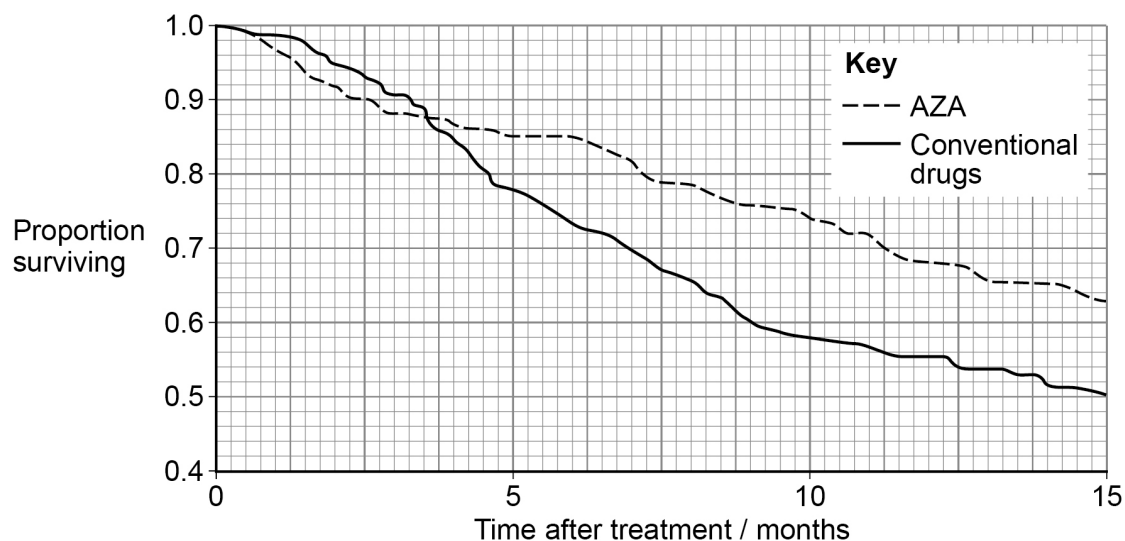
Suggest and explain how AZA can reduce the effects of MDS in some patients.

[3 marks]

Scientists investigated the effectiveness of AZA in patients with MDS.
A total of 360 patients were randomised in the ratio of 1:1 to receive AZA or
conventional drugs (control).

Figure 6 shows the scientists' results.

Figure 6



0 6 . 3

The control patients were treated with conventional drugs.

Give **two** reasons why.

[2 marks]

1 _____

2 _____

0 6 . 4

Use **Figure 6** and the information provided to calculate the difference in the number of patients surviving at 10 months after treatment with AZA compared with conventional drugs.

[2 marks]

Answer _____

